HAN JANG

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Seoul National University, 101 Daehak-ro, Jongno-gu, Seoul, South Korea
Healthcare AI · Agentic AI · Large Multimodal Models

Education

Integrated MS/Ph.D: *Mar. 2026 – Present*
Seoul National University – Seoul, South Korea
Interdisciplinary Program in Bioengineering, College of Engineering
Expected Graduation: 02/2031

Bachelor of Science: *Mar. 2019 – Aug. 2025*
Chungnam National University – Daejeon, South Korea
The Division of Computer Convergence,
College of Engineering

Research Interests

Multimodal Foundation Models
Cross-modal alignment, representation learning, and multimodal reasoning for healthcare.

Explainable and Trustworthy AI
Interpretable clinical decision-making, evidence-grounded reasoning, and robust medical AI systems.

Longitudinal Health Intelligence
Temporal representation learning, disease trajectory modeling, and counterfactual reasoning.

EXPERIENCE

Seoul National University Hospital (SNUH) – Seoul, South Korea ***Mar. 2025 – Present***
Researcher, Department of Radiology (AICON LAB, Prof. K.S. Choi)
Building agentic multimodal AI for healthcare, spanning domain-specific foundation models, diagnostic systems, and large-scale benchmarks validated in clinical practice.

NanoCellect Biomedical, Inc. – San Diego, CA, USA ***Apr. 2024 – Feb. 2025***
Machine Learning Engineer (Remote & On-site), R&D Team; with Lo Lab, UC San Diego
Applied ML/DL for cell image analysis (classification, segmentation, t-SNE) on VERLO™ Cell Sorter.

Seoul National University of Science and Technology – Seoul, South Korea ***Dec. 2023 – Aug. 2024***
Research Assistant, Dept. of Applied AI (BAIS LAB, Prof. K.H. Choi)
Developed bi-directional adversarial diffusion models for CT–MRI synthesis; published in ESWA (IF=9.4).

Korea Institute of Science and Technology (KIST) – Seoul, South Korea ***Jun. 2023 – Dec. 2023***
Research Intern, Bionics Research Center (AIMI LAB, Dr. K.H. Choi)
Built diffusion-based modality translation models and 3D segmentation frameworks for HCC and breast cancer.

Institute for Basic Science (IBS) – Daejeon, South Korea ***Apr. 2022 – Jun. 2023***
Research Assistant, Center for Cognition and Sociality (BrainX LAB, Dr. Y.J. Kim)
Conducted EEG experiments and built ML decoders for brainwave signal classification; modeled human visual cortex with predictive coding networks.

SELECTED PUBLICATIONS

Interpretable Multimodal Retrieval Augmented Diagnosis for Breast Ultrasound with Multinational Clinical Validation and Reader Study ***2026***
Han Jang*, S. An*, S. Jung, H. Ji, M. Kim, S. Lee, K. S. Choi, S. M. Ha, *npj Digital Medicine*, 2026, doi:10.1038/s41746-026-03272-3, (IF=18.0, JCR Top 0.52%).

MPIB: A Benchmark for Medical Prompt Injection Attacks and Clinical Safety in LLMs ***2026***
J. Lee*, Han Jang*, K. S. Choi, *Findings of EMNLP 2026*, Budapest, Hungary.

MedLayBench-V: A Large-Scale Benchmark for Expert-Lay Semantic Alignment in Medical VLMs *2026*
Han Jang*, J. Lee*, H. Eum, K. S. Choi, *Findings of ACL 2026*, San Diego, USA.

GlioSurv: Interpretable Transformer for Multimodal Survival Prediction in Diffuse Glioma *2025*
J. Lee, J. Jang, H. Eum, Han Jang, M. Kim, ..., K. S. Choi, *npj Digital Medicine*, 8(1), 660, 2025, (IF=18.0, JCR Top 0.52%).

Cyclic Conditional Diffusion Models for CT-to-MR Synthetic Image Segmentation with Misaligned Image Pairs *2026*
Han Jang*, N. Han*, J. Kwon, H. Seo, B. J. Park, K. Choi, *Expert Systems with Applications*, 304, 130631, 2026, (IF=9.4, JCR Top 5.8%).

PATENTS

Method and Apparatus for BI-RADS Classification of Breast Ultrasound Images *2026*
S. C. Lee, S. B. An, K. S. Choi, Han Jang, Korean Patent Application, filed May 28, 2026. (Pending)

ACADEMIC SERVICES

Reviewer, ***Int'l Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI 2026)*** *2026*

Reviewer, ***BMJ Digital Health & AI*** *2026–2027*

Reviewer, ***Frontiers in Physics*** *2026*

Reviewer, ***Frontiers in Oncology*** *2026*

TECHNICAL SKILLS

Machine Learning: PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, vLLM

Programming & Infrastructure: Python, C++, Multi-GPU Distributed Training (DDP), SLURM, Docker, Git

Medical Imaging: 3D Slicer, SimpleITK, ITK-SNAP, Elastix